

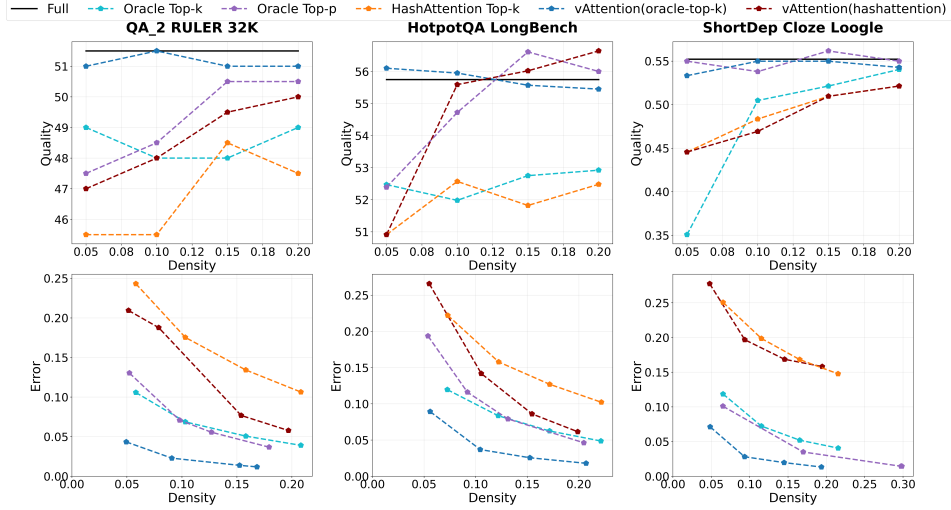


Figure 4: Pareto curves (Quality and Error vs. Density) for different baselines and their combination with vAttention across different datasets/benchmarks for Llama-3.1-8B-Instruct model. More pareto results are in Appendix A.1

$\epsilon \|N\|_2$  and  $\tau = \epsilon D$  in the numerator and denominator cases, respectively. Let  $b_D(\epsilon, \delta, \sigma, D, K, V, q)$  (resp.  $b_N(\epsilon, \delta, \text{Tr}(\Sigma), \|N\|_2, K, V, q)$ ) denote the minimum budget required to achieve an  $(\epsilon, \delta)$ -approximation for the denominator (resp. numerator). When parameters are clear from the context, we will drop those from the expression for convenience. The individual approximation results for the numerator and denominator can be combined to yield a bound on the quality of the approximated attention output, as stated in the lemma below.

**Lemma 4.2.** *If  $b_D$  and  $b_N$  are chosen such that we have  $(\epsilon_1, \delta_1)$  and  $(\epsilon_2, \delta_2)$  approximation on numerator and denominator respectively and  $\epsilon_2 < 0.5$ , then using  $b = \max(b_D, b_N)$  ensures that*

$$\Pr \left( \left\| \frac{N}{D} - \frac{\hat{N}}{\hat{D}} \right\|_2 > 2(\epsilon_1 + \epsilon_2) \right) \left\| \frac{N}{D} \right\|_2 < (\delta_1 + \delta_2) \quad (7)$$

**Comment.** The lemma above shows that if both the numerator and denominator are well approximated, then the overall attention output is also well approximated.

The results above can be combined into a single theorem providing an algorithm to select budget for  $(\epsilon, \delta)$  approximation of attention output.

**Theorem 4.3** ( $(\epsilon, \delta)$  **verified-SDPA**( $K, V, q$ )). *Let  $\Sigma$  be the covariance matrix for the population  $\{\exp \langle K[i], qV[i] \rangle\}_{i \in \mathcal{I}_f}$ . Let  $\sigma$  be the standard deviation for the population  $\{\exp \langle K[i], q \rangle\}_{i \in \mathcal{I}_f}$ . Then, if the budget is*

$$b \geq \min_{\epsilon' \in (0, \epsilon), \delta' \in (0, \delta)} \left[ \max \left( b_D \left( \frac{\epsilon'}{2}, \delta' \right), b_N \left( \frac{\epsilon - \epsilon'}{2}, \delta - \delta' \right) \right) \right] \quad \text{then} \quad (8)$$

$$\Pr(\|v\text{Attention}(K, V, q) - \text{SDPA}(K, V, q)\|_2 > \epsilon \| \text{SDPA}(K, q, v) \|_2) \leq \delta$$

Let  $b_{\text{SDPA}}(\epsilon, \delta, \Sigma\sigma, \|N\|_2, D, K, V, q)$  denote the minimum budget required for an  $(\epsilon, \delta)$  approximation of SDPA attention. When parameters are clear from the context, we will drop those in the expression for convenience. In practice,  $\Sigma$  and  $\sigma$  as well as exact  $\|N\|_2$  and  $D$  are not known apriori. However, we can use a base sample from residual tokens to estimate them in order to compute these quantities. In our algorithm, vAttention is parameterized by  $f_b$ , which denotes the fraction of  $n_s$  to be used as a sample for this estimation. The complete algorithms are provided in Algo. 1 and Algo. 2.

**Algorithmic Relaxations.** The exact computation of the budget specified in Equation 8, which can be performed entirely on GPU, can still be expensive. To address this, we adopt a relaxation that uses an  $(\epsilon, \delta)$ -approximation of the denominator alone. This choice is motivated by three key